Stochastic Process Assignment 2

Textbook Exercises

Chapter 3

52. Each applicant has a score. If there are a total of n applicants, then each applicant whose score is above s_n is accepted, where

$$s_1 = 0.2, \qquad s_2 = 0.4, \qquad s_n = 0.5 \quad \text{for } n \geq 3.$$

Suppose the scores of the applicants are independent uniform $(0, 1)$ random variables and are independent of N, the number of applicants, which is Poisson distributed with mean 2. Let X denote the number of applicants that are accepted. Derive expressions for

(a) $\mathbb{P}(X = 0)$.

(b) $\mathbb{E}[X]$.

Chapter 4

1. Three white and three black balls are distributed in two urns in such a way that each urn contains three balls. We say that the system is in state i, $i = 0, 1, 2, 3$, if the first urn contains i white balls.

At each step, we draw one ball from each urn and interchange the two balls: the ball drawn from the first urn is placed into the second urn, and the ball drawn from the second urn is placed into the first urn.

Let X_n denote the state of the system after the nth step. Explain why

$$\{X_n : n = 0, 1, 2, \ldots\}$$

is a Markov chain, and calculate its transition probability matrix.

2. Each individual in a population independently has a random number of offspring that is Poisson distributed with mean λ. Those initially present constitute the zeroth generation. The offspring of the zeroth-generation individuals constitute the first generation, their offspring constitute the second generation, and so on.

If X_n denotes the size of generation n, determine whether

$$\{X_n : n \geq 0\}$$

is a Markov chain. If it is, give its transition probabilities P_{ij}. If it is not, explain why not.

3. There are k players, with player i having value $v_i > 0$, $i = 1, \ldots, k$. In every period, two of the players play a game, while the other $k-2$ players wait in an ordered line. The loser of a game joins the end of the line, and the winner then plays a new game against the player who is first in line. Whenever players i and j play, player i wins with probability

$$\frac{v_i}{v_i + v_j}.$$

(a) Define a Markov chain that is useful in analyzing this model.

(b) How many states does the Markov chain have?

(c) Give the transition probabilities of the chain.

Additional Exercise

Provide two examples that can be modeled by a discrete-time Markov chain. For each example, define the state space and the transition probabilities.